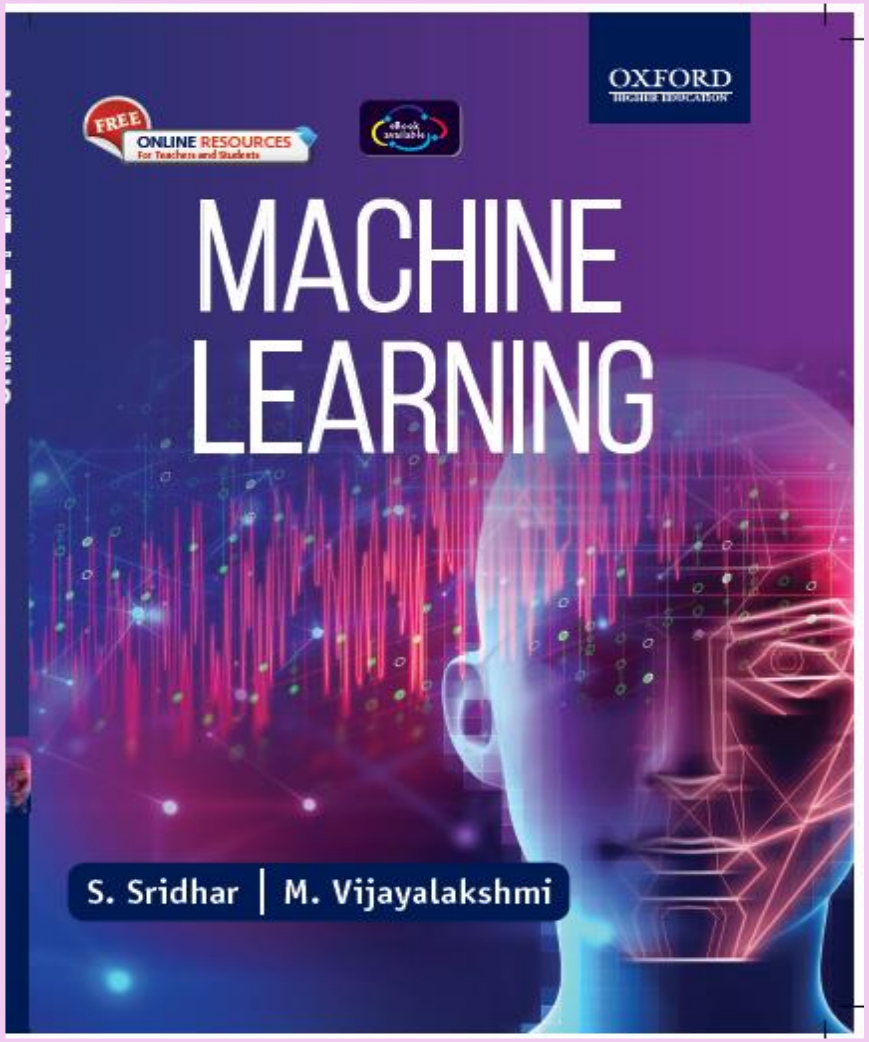


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Machine Learning

S. Sridhar and M. Vijayalakshmi

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Chapter 12

Ensemble Learning

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Ensemble Learning

Ensemble learning is a machine learning model which combines the predictions of several machine learning models to improve the prediction accuracy of using a single classifier. In other words, ensemble models have been developed to overcome the limitations of using a single-hypothesis machine learning model. The general principle of this model is to ensemble hypotheses of different models either by combining the collective predictions of different models into a single final model or by employing different models in a sequential fashion to increase the accuracy and robustness of learning. Ensemble models are useful to solve both classification and regression problems where they reduce bias and variance to boost the accuracy of models. They are used in many real-world applications, such as remote sensing, security, gene classification, image segmentation, video retrieval, healthcare, and so on.

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Introduction

Ensembling is a process of combining many weak learners to produce a strong learner that performs better than a single weak learner. The idea is to learn better with a set of models, instead of making one model to learn better. It is like getting opinions from many in order to decide and to protect oneself in decision-making.

A very simple way of ensembling is majority voting or taking average or weighted average of predictions of multiple machine learning models or base learners. 'Stacking' is another way of ensembling that uses the predictions of multiple models as 'features' to train a new model and then predict from that model. Another approach is to employ the same machine learning model multiple times with different datasets and combine the hypothesis generated from these models.

Bagging and Boosting are two popular models of ensemble learning. Bagging is a method of ensembling simultaneously many independent models and averaging the predictions obtained from the different models, which in fact, is more robust than the prediction made by a single classifier. Thus, bagging solves the overfitting problem which normally happens with a single classifier. On the other hand, boosting employs different classifiers in a sequential fashion where the prediction made by a classifier is enhanced by boosting the weights for the data instances and consecutively the predicted output is given as input to the next learning model. Thus, boosting increases the accuracy of predictions made by the previous learning model. Bagging minimizes the variance of the learning model, whereas boosting increases the accuracy of prediction.

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Advantages of Ensemble Learning

1. Minimizes bias and variance of a weak learner
2. Reduces overfitting and underfitting of the model
3. Increases accuracy of prediction
4. Handles higher dimensionality data efficiently

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Disadvantages of Ensemble Learning

1. Does not give precise values for the classification and regression model
2. Error prone and sometimes leads to overfitting
3. Requires careful tuning of the number and type of features used

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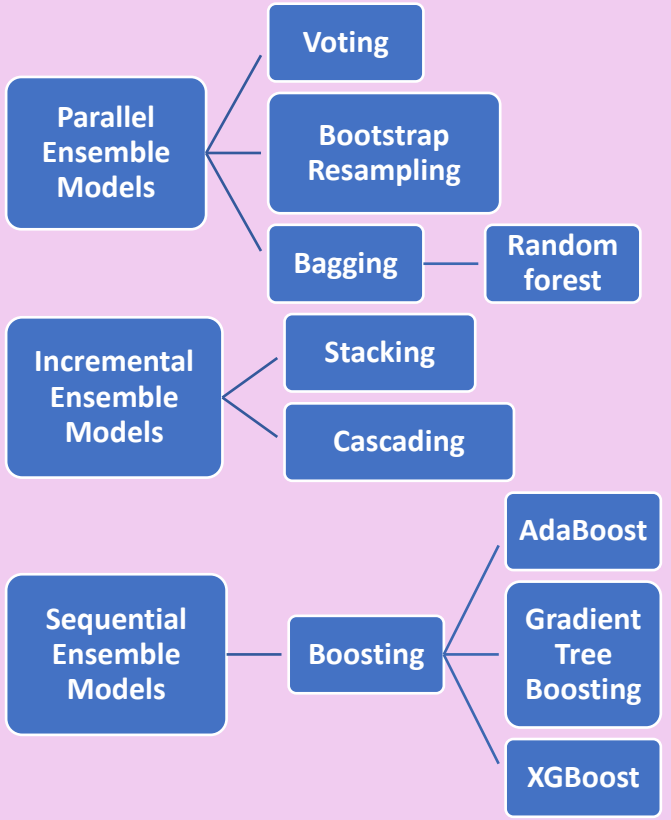
Ensembling Techniques

Ensemble models are generally classified as 'Parallel' ensemble models and 'Sequential' ensemble models based on how the base learners are ensembled for training. In a parallel ensemble mode, the base learners learn simultaneously and independently, whereas in a sequential ensemble mode the base learners are used sequentially to learn. Ensembling models can also be classified as 'Homogenous' and 'Heterogenous' ensemble models, depending on the type of base learners used in each stage. If same type of base learner is used for ensembling, then such a model is known as a Homogenous ensemble model, while in case of a heterogenous ensemble model, different type of base learner is used for ensembling.

The parallel ensemble models include Voting, Bootstrap Resampling and Bagging. Random forest model is a popular ensemble model constructed based on the idea of bagging. Some of the sequential ensemble models are AdaBoost, Gradient Tree Boosting and XGBoost which are also called as boosting models. Stacking and Cascading belong to the family of incremental ensemble models. Parallel ensemble learners are time and space efficient, as compared to sequential ensemble learners.

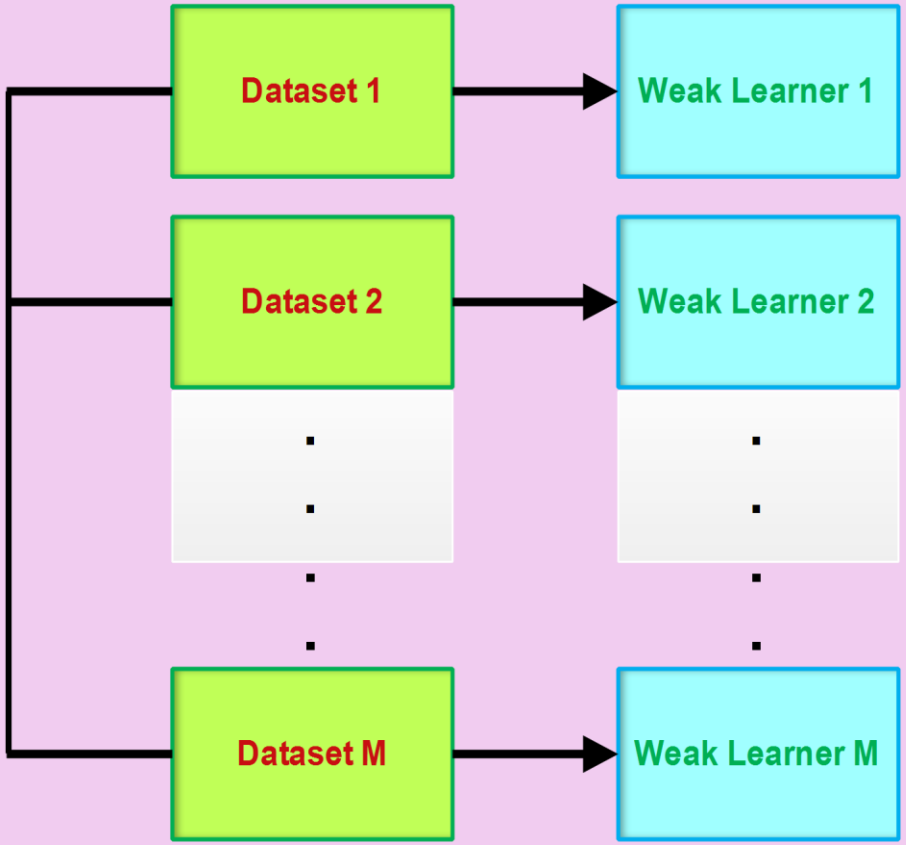
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Types of Ensemble Models



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Parallel Ensemble Models



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Parallel Ensemble Models

VOTING

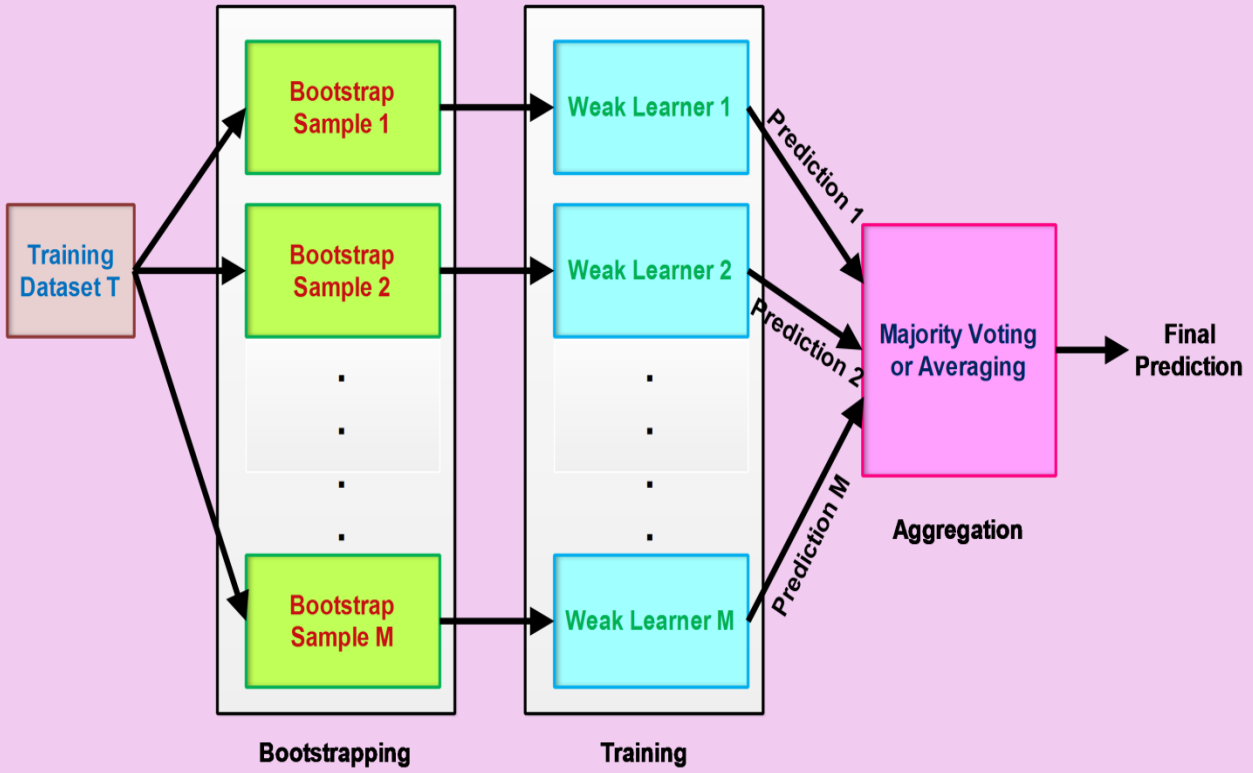
It is a simple way of ensembling multiple learners. It is the process of combining multiple machine learning models or using the same model multiple times as base learners and taking average of their predictions in the case of regression problem and considering the maximum vote of their predictions in the case of classification problem.

BOOTSTRAP RESAMPLING

It is a process of creating many bootstrap samples $S_i (1 \leq i \leq M)$ using a method called Random Sampling with replacement from the training dataset. Each bootstrap sample contains 63.2% distinct tuples from the original training dataset and the remaining tuples are duplicate tuples drawn from the original dataset. If the training dataset T contains N tuples, M samples can be created, where each bootstrap sample S_i consists of N tuples by randomly choosing N tuples from the original training dataset with replacement.

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Bagging



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Bagging

Algorithm 12.1: Bagging

Input: Training dataset T with N data instances.

Step 1: Create M bootstrap samples $S_i (1 \leq i \leq M)$ of N data instances each from the training dataset T using Random sampling with Replacement.

Step 2: Train M weak classifiers $H_i (1 \leq i \leq M)$ using the samples $S_i (1 \leq i \leq M)$.

Step 3: Combine predictions from all classifiers taking average $H_{ave} = \frac{1}{N} \sum_{i=1}^{M} H_i$.

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Random Forest

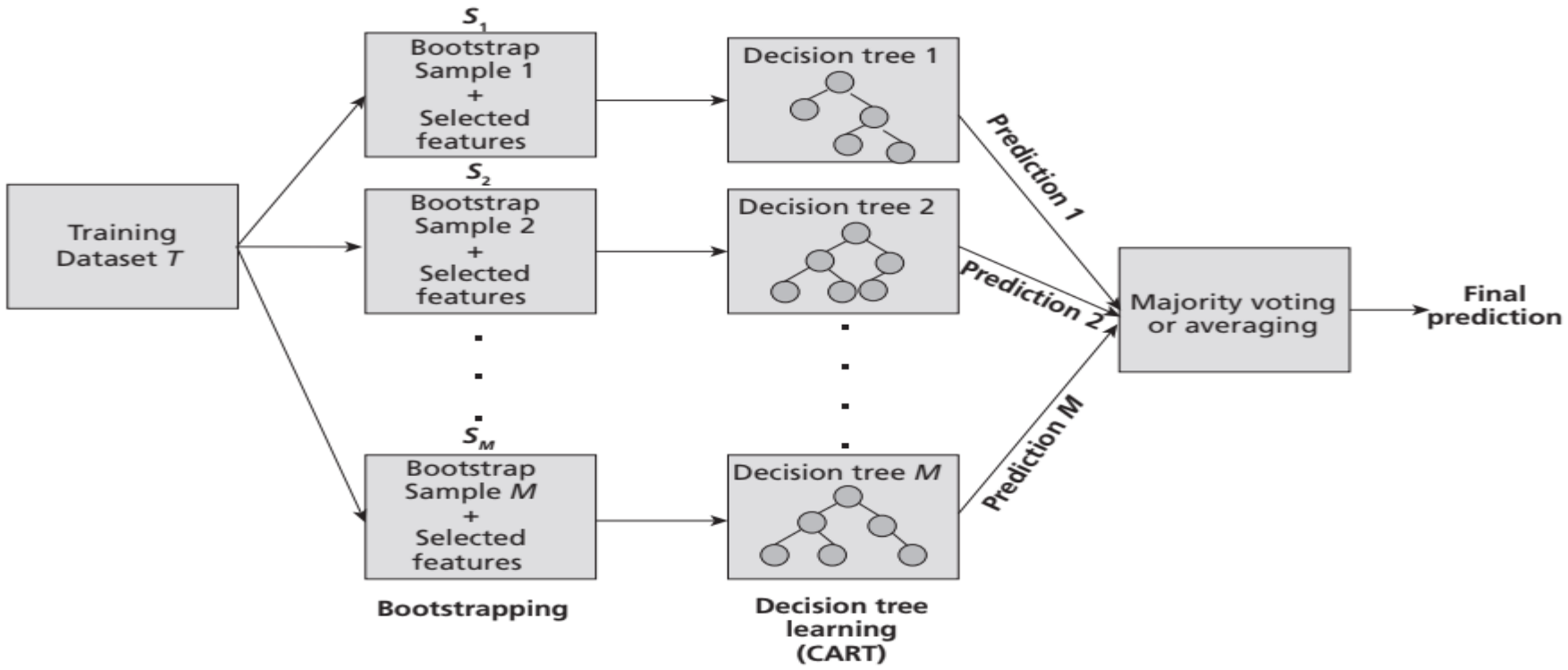


Figure 12.1: Random Forest Model

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Random Forest

Algorithm 12.2: Random Forest

Input: Training dataset T with N data instances.

Step 1: Create M bootstrap samples $S_i (1 \leq i \leq M)$ of N data instances each from the training dataset T using Random sampling with Replacement.

Step 2: Choose number of Decision trees to be constructed DT_M.

Step 3: Choose the number of features to be used in each tree. Normally, $\sqrt{D}$ randomly chosen features are taken.

Step 4: Choose the depth of the decision tree. All trees are to be same depth.

Step 5: Repeat M times. For each tree:

(a) Choose a bootstrap sample S_i of N data instances.

(b) Compute the best split feature to be placed at each node.

(c) Construct decision tree using CART. Do not prune the tree.

Step 6: Combine predictions from all classifiers taking Average $H_{ave} = \frac{1}{N}\sum_{i=1}^{M} H_i$ for classification problem or Majority voting in the case of regression problem.

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Random Forest

ADVANTAGES OF THE RANDOM FOREST MODEL

1. Can solve both classification problems and regression problems
2. Reduces overfitting and variance compared to a single decision tree
3. Provides more accuracy
4. Performs well even with smaller set of data

DISADVANTAGES OF THE RANDOM FOREST MODEL

1. Increased complexity in construction and prediction
2. Uses more computational resources

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Incremental Ensemble Models

Incremental ensemble models use the predictions that are learnt from the previous learner as features to train a meta learner and make final predictions.

STACKING
Stacking model includes two types of learners called base-learners and meta-learners.

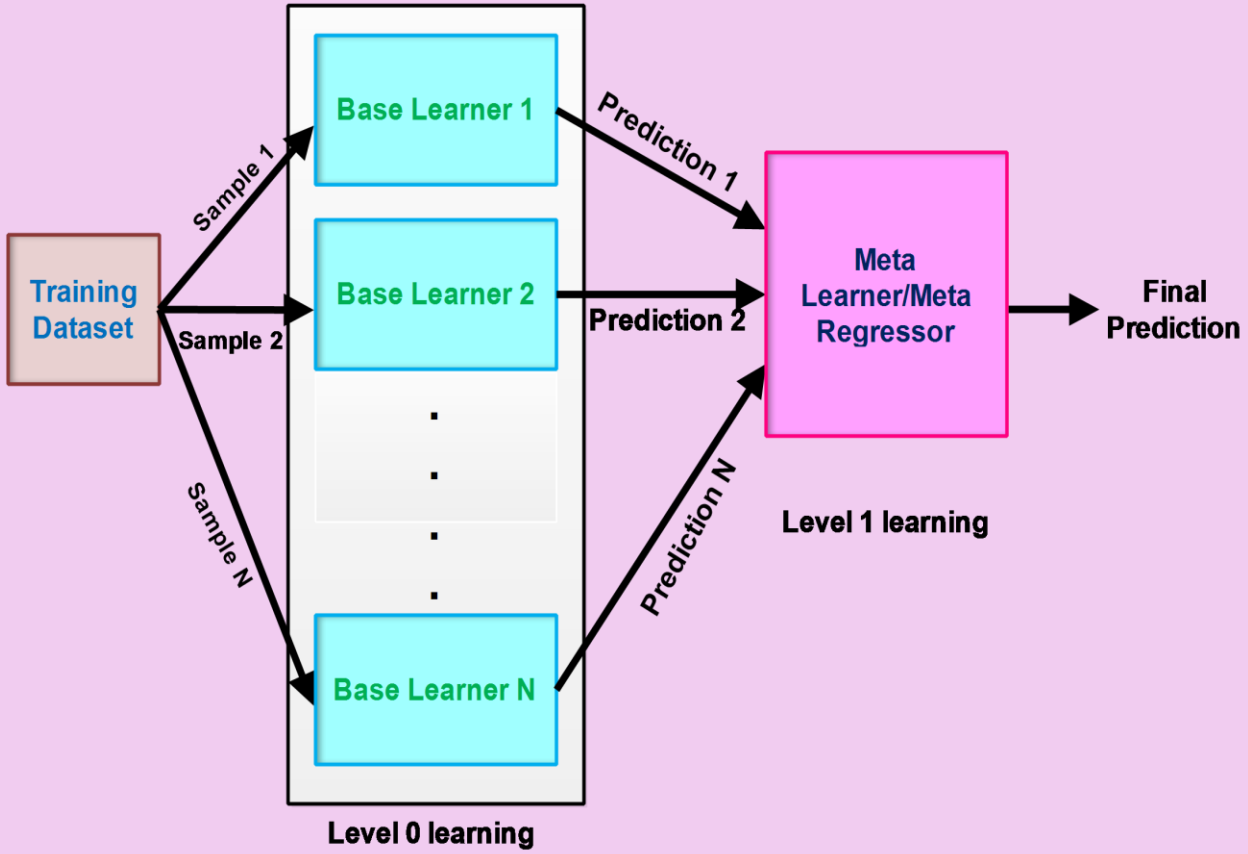
It is a process of ensembling multiple heterogeneous machine learning models or base learners to make predictions and then use these predictions as features to train the second level meta learner model whose prediction is considered as the final prediction.

CASCADING
A cascading ensemble model is a multistage learning technique with a sequence of classifiers ordered in terms of increasing complexity and specificity such that early classifiers are simple and general whereas later ones are more complex and specific.

In cascaded models, different classifiers are coupled with their input/output function in a cascade, improving performance at each level.

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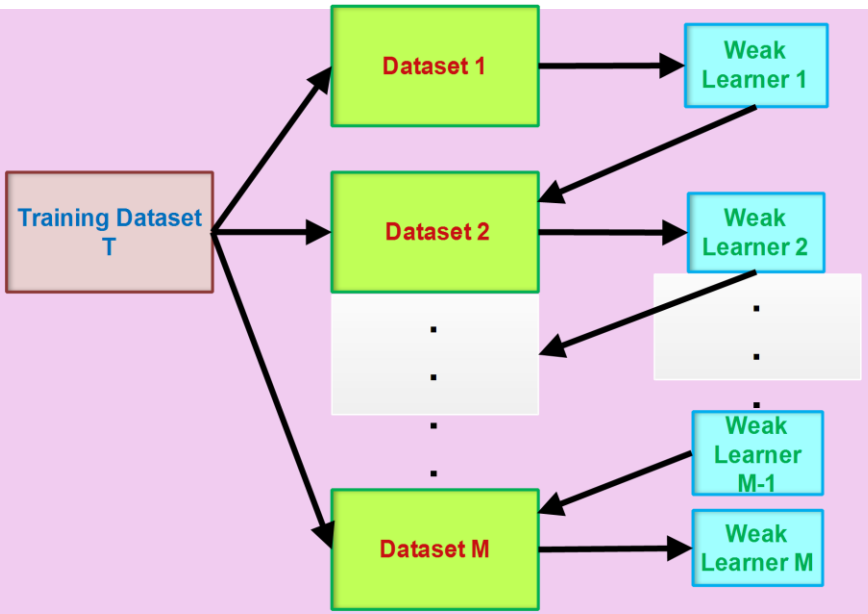
Stacking



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Sequential Ensemble Models

Boosting algorithms such as AdaBoost, Gradient Boosting and XGBoost are categorized as Sequential Ensemble models. The general principle of this model is to use multiple weak learners sequentially and the error exhibited by a weak learner is used to give importance to data instances which are further retrained by the next weak learner. Thus, the accuracy of prediction is improved in the ensemble model.



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AdaBoost

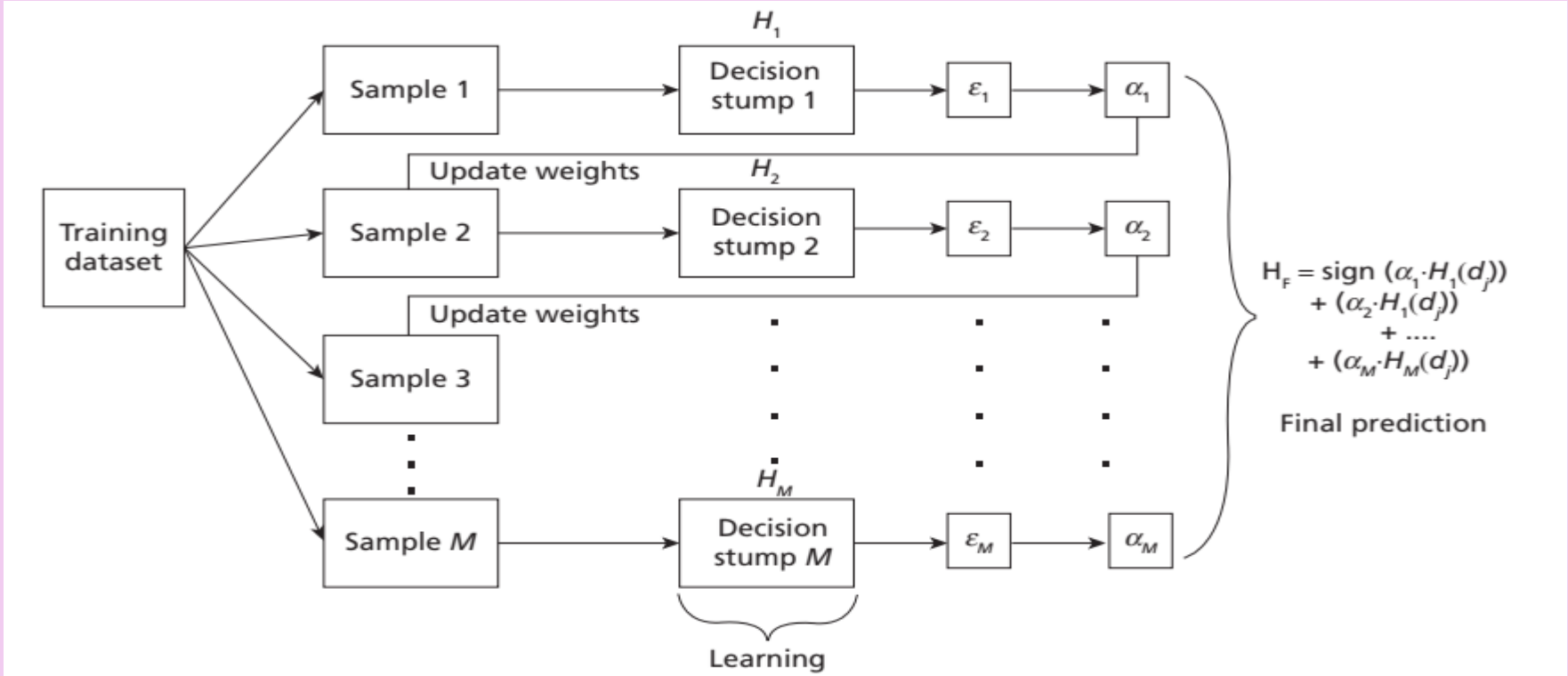


Figure 12.5: Working of AdaBoost Algorithm

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Algorithm 12.3: AdaBoost

Algorithm 12.3: AdaBoost

Input: Training dataset T with N data instances and M weak classifiers.

Step 1: Assign uniform weights 1/N to all the data instances d_j $(1 \le j \le N)$ in the training dataset T, i.e., $wt\ (d_1) = 1/N$, $wt\ (d_2) = 1/N$, …, $wt\ (d_N) = 1/N$. Here, wt implies weight.

Step 2: Repeat for each weak classifier.

Step 2 (a): Train a weak classifier H_i with a random bootstrap sample from the training dataset T.

Step 2 (b): Compute the weighted error ε_i of H_i on current training dataset:

$$\varepsilon_i = \sum_{j=1}^{N} H_i(d_j) wt(d_j)$$

$H_i(d_j) = 0$ if prediction is correct with H_i

$H_i(d_j) = 1$ if prediction is wrong with H_i

Step 2 (c): Compute the weight of each weak classifier:

$$\alpha_i = \frac{1}{2} \ln\left(\frac{1-\varepsilon_i}{\varepsilon_i}\right)$$

α_i is based on the error rate.

As the error rate increases, α_i grows exponentially negative.

Better classifiers get exponentially more weight.

If error rate is 50%, the classifier gets a weight of 0.

If error rate is more than 50%, the classifier gets a negative weight.

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Algorithm 12.3: AdaBoost

Step 2 (d): Calculate the normalizing factor Z_i. This is needed so that the sum of the weights of data instances adds up to 1.

Z_i = Total weight of correctly classified instances + Total weight of incorrectly classified instances and Here, *wt* implies weight.

$$Z_i = wt(\text{correctly classified instance}) \times \text{Number of correct classifications} \times e^{-\alpha_i} + wt\begin{pmatrix}\text{incorrectly}\\ \text{classified instance}\end{pmatrix} \times \text{Number of incorrect classifications} \times e^{+\alpha_i}$$

Step 2 (e): Update the weight of all data instances:

$$wt(d_j)_{i+1}\ \text{correct-instance} = \frac{wt(d_j)_i\ \text{correct} - \text{instance} \times e^{-\alpha_i}}{Z_i}$$

$$wt(d_j)_{i+1}\ \text{incorrect-instance} = \frac{wt(d_j)_i\ \text{incorrect} - \text{instance} \times e^{+\alpha_i}}{Z_i}$$

where, $wt(d_j)_{i+1}$ correct-instance, is the updated weight of a correctly classified data instance to be trained by the next weak classifier.

$wt(d_j)_{i+1}$ incorrect-instance, is the updated weight of an incorrectly classified data instance to be trained by the next weak classifier.

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Algorithm 12.3: AdaBoost

Data instances that are wrongly classified will have their weights increased.

Data instances that are classified correctly will have their weights decreased.

Step 3: Compute the final predicted value for each data instance. It is the linear combination of weighted average predictions of all the classifiers multiplied by weight α_i computed for each classifier.

$$H_F(d_1) = \text{sign}\left(\sum_{i=1}^{M} \alpha_i \times H_i(d_j)\right)$$

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AdaBoost

ADVANTAGES OF ADABOOST ALGORITHM

1. FAST

2. SIMPLE TO IMPLEMENT

DISADVANTAGES OF ADABOOST ALGORITHM

1. FROM EMPIRICAL EVIDENCE AND THEORETICAL EVIDENCE, ADABOOST MAY OVERFIT

2. SENSITIVE TO NOISE DATA AND OUTLIERS

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Gradient Tree Boosting

Gradient tree boosting is also known as gradient boosting machine (GBM) or gradient boosted regression tree (GBRT).

Gradient boosting is a combination of two procedures called gradient descent and boosting.

Gradient boosting or gradient tree boosting is also a boosting algorithm which employs weak learners in a sequential fashion but optimizes the loss function or minimizes the error at each stage by moving in the opposite direction of the gradient to reach the minima.

The algorithm boosts the residual errors made by a weak learner rather than boosting the weights of misclassified data instances as done by the ADABOOST algorithm.

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XGBoost

XGBOOST stands for extreme gradient boosting.

It is a tree ensemble model that belongs to the family of gradient boosting decision tree (GBDT) models.

It is a scalable tree boosting system that combines both hardware optimizations and algorithmic optimizations to provide computational speed and high performance with optimized usage of resources.

The boosting model proves an improvement by adding regularization with gradient boosting to avoid overfitting and bias.

This model provides an open end-to-end tree boosting scalable system.

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Features of XGBoost

REGULARIZATION

IN-BUILT CROSS VALIDATION

POST PRUNING

IT ADDS TWO **ALGORITHMIC OPTIMIZATIONS** TO GRADIENT BOOSTING CALLED,

WEIGHTED APPROXIMATE QUANTILE SKETCH

SPARSE AWARE ALGORITHM

IT ALSO SUPPORTS **HARDWARE OPTIMIZATIONS** SUCH AS,

OUT OF CORE COMPUTING

PARALLELIZATION

CACHE AWARENESS

DISTRIBUTED COMPUTING

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Summary

1. Ensemble learning is a machine learning model which combines the predictions of several classifiers in order to improve the prediction accuracy of using a single classifier.
2. Bagging and Boosting are two popular models of ensemble learning.
3. A simple way of ensemble learning is majority voting or taking an average or weighted average of predictions of multiple machine learning models or base learners.
4. Ensembling machine learning is an approach to minimize bias and variance, and to build accurate models that avoid over-fitting and under-fitting of learning models.
5. Bagging is otherwise called as bootstrap aggregation, which combines both bootstrapping and aggregation.
6. The random forest ensemble model is a variant of Bagging which uses decision tree for learning.
7. Boosting algorithms train multiple weak learners sequentially and the error exhibited by a weak learner is used to give importance to data instances which are further retrained by the next weak learner.
8. AdaBoost, called as adaptive boosting, is a popular boosting algorithm that trains multiple weak classifiers sequentially so that the errors made by poorly performing weak classifiers are reduced when they are trained sequentially.
9. AdaBoost is a greedy algorithm optimizing weights of data instances in the training dataset by adding a weak classifier at each step.

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